Meriem JERBI

Data Science Engineer
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Profile

Data Science Engineer graduated from ENET'Com Sfax, specialized in **Machine Learning, Deep Learning, NLP, and Generative AI**. Experienced in the full model lifecycle: data preparation, **fine-tuning of LLMs** and **GNNs**, rigorous evaluation, as well as building **RAG systems**. Strong skills in **Python**, **PyTorch**, and **Hugging Face**. Seeking a Data Scientist / Machine Learning Engineer position to turn data into reliable, high-impact solutions in a collaborative, fast-paced team.

Education

National School of Electronics and Telecommunications of Sfax (ENET'Com)
Engineering Degree – Data Engineering and Decisional Systems *2023 – 2026*

Faculty of Sciences of Sfax (FSS)
Preparatory Cycle in Mathematics and Physics (MP) *2021 – 2023*

Lycée Tayeb Mhiri – Sfax
Baccalaureate in Mathematical Sciences *2021*

Professional Projects

Predicting Material Synthesizability with LLMs and Graph Neural Networks
Final Year Internship (PFE) – UCCS, University of Artois (Lens, France)

- Built an end-to-end AI pipeline to filter AI-generated candidate crystal structures by synthesizability, reducing costly DFT calculations downstream in a materials-discovery project.
- Fine-tuned **Llama-3.1-8B, Qwen2-7B, and Gemma-2-9B** with **LoRA** on ∼100k crystal structure descriptions for binary classification, and extended the approach to **explanation generation** and **precursor prediction**.
- Fine-tuned and compared four **Graph Neural Networks** (**ALIGNN, CHGNet, M3GNet, MEGNet**) using partial layer freezing and a custom classification head.
- Designed an **LLM–GNN fusion model** (4096-d Llama embeddings + 256-d ALIGNN embeddings), reaching **96.96% accuracy** and **0.997 ROC-AUC**.
- Ran experiments on a **GPU cluster (SLURM, Linux)** and collaborated through **Git/GitHub** actions.
- **Technologies:** Python, PyTorch, Hugging Face (Transformers, PEFT), LoRA, pymatgen, scikit-learn, SLURM, Git.

PsyBot – RAG-based Conversational Chatbot
Summer Internship – ReGIM-Lab, Enis (Sfax, Tunisia)

- Design and development of an intelligent chatbot based on **Retrieval-Augmented Generation (RAG)** to provide contextual and personalized responses.
- Implementation of a **hybrid retrieval approach** combining **BM25** and **vector search (Qdrant)**, with a **reranking** mechanism.
- Generation of semantic embeddings using **BGE-M3** to improve response relevance.
- Integration of **LLMs via Ollama** within a scalable web architecture.
- **Technologies:** React.js, Node.js, Firebase, Ollama, Qdrant, BM25, BGE-M3.

Academic and Personal Projects

Spam Detection System (NLP & Deep Learning)

- Development of an **LSTM model** for spam message classification achieving **98% accuracy**.
- Advanced text preprocessing and hyperparameter optimization.
- **Technologies:** Python, TensorFlow/Keras, NLP, NLTK.

SAVI-AI – Arabic Violent Speech Detection

- Design of a text classification system for Arabic content into *violent, non-violent, or neutral.*
- Training and comparison of **Naïve Bayes** and **AraBERT** models.
- Integration of a reactive chatbot providing real-time feedback to users exhibiting harmful linguistic patterns, promoting positive communication.
- **Technologies:** Python, TensorFlow, AraBERT, LLMs, Flask, Flutter.

Sahha AI – Medical NLP Diagnostic System for Tunisian Dialect

- Design of an NLP system based on **Transformers** to translate Tunisian dialect (Derja / Arabizi) into standard medical French.
- Implementation of a **vector semantic inference** approach to estimate disease probabilities from symptoms.
- Use of **Gemini 2.5 Flash** and comparison with **GPT-4, Claude, and Qwen** in terms of linguistic robustness, latency, and cost.
- System evaluation using quantitative metrics such as **F1-score**, **Top-1 Accuracy**, and **BLEU Score**.
- **Technologies:** Transformers, LLMs, NLP, Embeddings, Cosine Similarity.

Technical Skills

Programming: Python, SQL, Java, C, JavaScript, PHP, Scala
Machine Learning & Deep Learning: Supervised Learning, Classification, Model Evaluation, Hyperparameter Tuning, LSTM, Graph Neural Networks.
NLP & Generative AI: Transformers, LLM Fine-Tuning (LoRA / PEFT), RAG, Embeddings, Hybrid Search (BM25 + vector), Prompt Engineering, Ollama.
Libraries & Frameworks: PyTorch, TensorFlow/Keras, Hugging Face (Transformers, PEFT), scikit-learn, Pandas, NumPy, NLTK, Flask.
Databases & Vector Stores: MySQL, MongoDB, Firebase, Qdrant.
Tools & Environments: Linux, HPC / SLURM, Git & GitHub, Jupyter Notebook, Google Colab, VS Code.
Web & Mobile: React.js, Node.js, Flutter.

Certifications and Training

- Microsoft Azure AI Fundamentals
- Python & Machine Learning Training
- Python / SQL / SQLite Training
- Generative AI and Computer Vision Workshop
- Flutter Development Workshop
- Participation in a Web Development Hackathon

Languages

Arabic (Native)
English (TOEIC Certified)
French (B1)
German (A2)